Reservoir Engineer, Modeling

Fervo Energy · Golden, CO

Job ID (this archive)	ENERGY-112
Employer	Fervo Energy
Location(s)	Golden, CO
Career stage (archive tag)	Experienced (Mid-career)
Location group	Other U.S.
Majors (archive tags)	Mechanical Engineering, Data Science, Computational & Geospatial Science
Energy area	Geothermal / geospatial and subsurface modeling
Pay listed	$140,000 – $200,000 per year
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/fervoenergy/jobs/4430122009
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-083

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Fervo Energy

Fervo Energy (NASDAQ: FRVO) is a modern power company developing next-generation geothermal resources to meet the growing demand for affordable, reliable, 24/7 carbon-free energy. Our mission is to transform geothermal into America's most dependable and affordable source of clean, around-the-clock power. By combining innovations in horizontal drilling, fiber-optic sensing, and advanced reservoir engineering, Fervo has developed a repeatable approach to building utility-scale enhanced geothermal systems. Our flagship Cape Station development in Utah is planned to deliver 500 MW of firm, carbon-free power, and we are advancing a development pipeline of more than 50 GW. Join us and work alongside world-class engineers, geoscientists, operators, project builders, and business leaders working to make geothermal a cornerstone of a more reliable, affordable, and sustainable energy future.

Position Description

Joining Fervo as a Reservoir Engineer focused on reservoir modeling and history matching means owning the calibration of our subsurface models against real field performance — both the stimulation fracture growth created during completions and the production and pressure response of the reservoir over time. This role also defines and drives the testing and data acquisition programs needed to close calibration gaps and continuously improve history-matching accuracy. The position is ideal for an engineer who enjoys rigorous model-to-data reconciliation, diagnosing the sources of mismatch, and designing the next test that resolves it.

The Reservoir Engineer will build, calibrate, and maintain reservoir and fracture models, with a strong focus on history matching both stimulation-induced fracture growth and reservoir production/pressure behavior. This role emphasizes quantitative rigor, root-cause diagnosis of model mismatch, and the design of targeted diagnostic testing to improve match quality over time.

The position collaborates closely with Completions, D&C, Reservoir, and Data Science teams to align stimulation design, field data acquisition, and simulation workflows so that models stay predictive as the asset base grows.

Responsibilities

Stimulation & Fracture Growth History Matching

- Build and calibrate hydraulic fracture growth models (e.g., ResFrac or comparable geomechanical/fracture simulators) against observed stimulation data.
- History-match fracture height, length, and complexity using treatment pressure, microseismic, DAS/DTS, and tracer data.

- Characterize in-situ stress state and natural fracture (NF) interactions using fracture mechanics principles to guide fracture growth and stress-shadow modeling.
- Account for fracturing-fluid rheology in fracture models, including its influence on fracture width, height growth, and proppant transport.
- Incorporate completion design parameters (stage spacing, cluster design, pump schedule, fluid/proppant systems) into fracture growth models and iterate as new stimulation data arrives.
- Translate calibrated fracture models into simulation-ready fracture network representations for downstream reservoir modeling.
- Quantify uncertainty in fracture geometry and connectivity and communicate its impact on well and pad performance predictions.

Reservoir Production & Pressure History Matching

- Build and maintain reservoir simulation models (full-field and sector-scale) representing fractured EGS reservoirs.
- History-match production and injection rates, bottom-hole pressures, and thermal/temperature response against field data.
- Iterate reservoir model parameters (permeability, fracture conductivity, compartmentalization, injector-producer connectivity) to improve match quality across wells and pads.
- Diagnose root causes of history-match mismatch and distinguish model, data, and operational sources of error.
- Use history-matched models to inform production forecasts, well spacing, and injection/production strategy.

Testing & Data Program Development

- Design diagnostic and surveillance testing programs — pressure transient tests, interference tests, tracer tests, DFITs, PLT/DTS/DAS — specifically to improve history-matching accuracy.
- Identify the data gaps most limiting model calibration and prioritize acquisition programs to close them.
- Define minimum viable data requirements per well and pad to support robust history matching within cost and schedule constraints.
- Continuously evaluate testing protocols against realized model performance and refine the testing program based on lessons learned.
- Extend calibrated models beyond history matching to new prospects, forecasting anticipated crossflow behavior and helping design corresponding crossflow test programs; scope and cadence for this forward-looking work will be defined in coordination with Resource and Exploration teams.
- Build repeatable workflows (including automation/scripting) to speed up history-matching iteration and testing feedback loops.

Cross-Functional Collaboration

- Partner with Completions and D&C to align stimulation design changes with fracture model predictions and observed outcomes.
- Partner with Reservoir and Resource teams to ensure production, injection, and pressure data flows support ongoing model calibration.
- Document modeling methodologies and calibration standards to promote consistency across projects.

Team and Culture

- Operate with technical independence and objectivity while maintaining strong collaborative relationships with Completions, D&C, and asset teams.
- Provide constructive challenge and thoughtful recommendations grounded in data and engineering principles.
- Balance rigor with pragmatism in a fast-paced, cross-functional environment.
- Contribute to a culture of transparency, consistency, and continuous improvement across the portfolio.
- Be responsive and reliable in remote settings, maintaining momentum without requiring constant oversight.
- Adapt quickly to shifting priorities, new data, and evolving model scopes.

Qualifications

Required

- B.S. in Engineering (Petroleum, Mechanical, Chemical, or related discipline).
- 5+ years of experience in reservoir engineering, reservoir simulation, or hydraulic fracture modeling; advanced degree may be considered in lieu of experience.
- Hands-on experience with reservoir and/or fracture simulation software (e.g., ResFrac, CMG, Eclipse, or comparable).
- Experience history matching production, pressure, and/or fracture growth data against simulation models.

- Strong understanding of fluid flow in fractured/porous media and hydraulic fracture mechanics.
- Experience interpreting pressure transient and completion diagnostic data (DFITs, PLT/DTS/DAS, tracer, microseismic).
- Strong analytical and programming/scripting skills (e.g., Python) to support model calibration and workflow automation.
- Strong analytical, synthesis, and communication skills, including the ability to explain model uncertainty clearly.

Preferred

- Experience in geothermal reservoir engineering, enhanced geothermal systems (EGS), or unconventional resource development.
- Direct experience with ResFrac or other coupled geomechanics/fracture-reservoir simulators.
- Working knowledge of fracture mechanics and in-situ stress state, including natural fracture (NF) interactions, to guide fracture growth and stress-shadow modeling.
- Familiarity with fracturing-fluid rheology and its impact on fracture geometry, height growth, and proppant placement.
- Experience designing diagnostic testing programs specifically to support model calibration, including forward-looking crossflow test design for new prospects.
- Experience integrating multiple diagnostic datasets (DAS/DTS, tracer, microseismic, PLT) into a single calibration workflow.
- Experience with assisted/automated history matching techniques (e.g., ensemble methods, machine learning-assisted calibration).
- Ability to translate detailed technical analysis into executive-level summaries.

Location

Fervo Energy has offices in Houston, TX, Golden, CO, Reno, NV, Oakland, CA, and Salt Lake City, UT. This position will be eligible for some hybrid work flexibility, but regular in-office presence at any of the above locations will be required.

Compensation

Fervo provides a comprehensive suite of benefits including medical, dental, vision, life, short-term and long-term disability, flexible paid time off, and paid parental leave. Additionally, Fervo offers an incentive stock options program, a bonus incentive program, and a 401(k) plan with an employer match.

Fervo Energy is providing the compensation range and general description of other compensation and benefits that the company in good faith believes it might pay and/or offer for this position based on the successful applicant’s education, experience, knowledge, skills, and abilities in addition to internal equity and geographic location. Expected Salary: $140,000-$200,000 based on Colorado locality, pay in other locations may vary.

Fervo Energy reserves the right to ultimately pay more or less than the posted range and offer other compensation, depending on circumstances not related to an applicant’s sex or other status protected by local, state, or federal law.

Benefits & Perks

- Fervo offers a comprehensive benefits package designed to support your health, financial well-being, and life outside of work. Benefits for eligible employees include:
- Medical, dental, and vision plan options
- Employer-paid basic life insurance and short-term and long-term disability coverage
- A 401(k) plan with up to 6% match
- Flexible paid time off and paid parental leave
- Family-building, fertility, hormonal health, pregnancy, postpartum, and parenting support through Carrot
- A dependent care FSA for eligible child and elder care expenses
- An annual Lifestyle Spending Account for eligible wellness expenses, including fitness, financial planning, pet insurance, and pet care
- Employee ownership opportunities through Fervo’s Employee Stock Purchase Program, as well as LTI and bonus opportunities for eligible roles

Benefits and programs are subject to eligibility requirements and may change.

Our Workplace

Fervo Energy highly values having employees work in the office to foster collaboration and strengthen our company culture. This role follows a hybrid schedule, with team members expected to work in the office three days per week. Fervo considers this in-office schedule an essential function of the role. Your recruiter can share more information about the schedule.

Equal Opportunity Employer

Fervo Energy is an Equal Opportunity Employer and does not discriminate on the basis of race, color, creed, gender, religion, marital status, registered domestic partner status, age, national origin, ancestry, physical or mental disability, medical condition, sex, genetic information, sexual orientation, military or veteran status, or any other consideration made unlawful by federal, state, or local law. Fervo also prohibits discrimination based on the perception that an individual has any of these characteristics or is associated with a person who has or is perceived as having any of these characteristics.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: B.S. in Engineering (Petroleum, Mechanical, Chemical, or related discipline). | B.S. in Engineering (Petroleum, Mechanical, Chemical, or related discipline). | 5+ years of experience in reservoir engineering, reservoir simulation, or hydraulic fracture modeling; advanced degree may be considered in lieu of experience.

Key skills listed in the archive: Python; machine learning; forecasting; automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 4430122009

Career stage: Mid-career

Major fit: Direct geospatial/computational pathway; exact education and experience requirements retained. Energy-sector connection: GIS, seismic analysis or subsurface simulation for geothermal electricity generation.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 4430122009.

Sources and retrieval details

Employer posting

https://job-boards.greenhouse.io/fervoenergy/jobs/4430122009

Retrieved source

https://boards-api.greenhouse.io/v1/boards/fervoenergy/jobs?content=true

Retrieved: 2026-10-26T16:33:09.471773+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.